

Viva Q&A by Manvendra

Lecture 1 — Introduction to Generative AI & Foundation Models

Q1: What was the significance of AlexNet in deep learning history?

A: AlexNet achieved a breakthrough in the ImageNet challenge, demonstrating the power of deep learning for image classification.

Q2: What is the biological inspiration behind deep learning? A: Deep learning is inspired by the structure and functioning of biological neurons in the brain.

Q3: Name two real-world AI applications mentioned in Lecture 1 besides AlexNet. A: AlphaGo (game-playing AI) and Visual Microphone (extracting sound from visual vibrations).

Lecture 2 — Generative Models & GANs

Q1: What is the key difference between generative and discriminative models? A: Discriminative models learn the decision boundary $P(y|x)$, while generative models learn the data distribution $P(x)$ to generate new samples.

Q2: What are the two components of a GAN and their roles? A: Generator creates fake samples to fool the discriminator; Discriminator distinguishes real samples from fake ones.

Q3: What is the training objective of a GAN? A: A minimax game where the generator minimizes and the discriminator maximizes the same loss, converging at Nash Equilibrium.

Q4: Name three common challenges in GAN training. A: Mode collapse, training instability, and vanishing gradients.

Q5: What is the connection between GAN loss and Jensen-Shannon Divergence (JSD)? A: The GAN minimax loss is related to minimizing the Jensen-Shannon Divergence between the real and generated data distributions.

Lecture 3 — Autoencoders & VAEs

Q1: What is the main problem with a standard autoencoder's latent space? A: The latent space is discontinuous — nearby points may not decode to similar outputs, making it unsuitable for generation.

Q2: How does a VAE differ from a standard autoencoder in encoding?

A: A VAE encodes input to a distribution (mean and std) instead of a single point, ensuring a smooth latent space.

Q3: What is the reparameterization trick in VAEs and why is it needed? A: $z = \mu + \sigma \cdot \epsilon$ (where $\epsilon \sim N(0,1)$); it makes sampling differentiable so gradients can flow through during backprop.

Q4: What are the two components of the VAE loss (ELBO)? A: Reconstruction loss (how well the output matches input) + KL divergence (how close the learned distribution is to a standard normal).

Lecture 4 — Transformers & Attention

Q1: Why are Transformers preferred over RNNs? A: Transformers allow parallelization and handle long-range dependencies better than sequential RNNs.

Q2: What are Q, K, V in the attention mechanism? A: Query (what I'm looking for), Key (what I contain), Value (what I return if matched) — used to compute weighted attention scores.

Q3: Why do we scale by $\sqrt{d_k}$ in scaled dot-product attention? A: To stabilize variance of dot products — without scaling, large d_k causes extreme softmax values (vanishing gradients).

Q4: What is the purpose of causal masking in Transformers? A: It prevents the model from attending to future tokens during autoregressive generation (e.g., in GPT).

Q5: What is multi-head attention? A: Running multiple attention operations in parallel with different learned projections, then concatenating their outputs.

Lecture 5 — Self-Attention vs Cross-Attention

Q1: What is the difference between self-attention and cross-attention? A: In self-attention, Q/K/V come from the same sequence; in cross-attention, Q comes from one source and K/V from another.

Q2: Where are MHSA and MHCA placed in the U-Transformer architecture? A: MHSA is placed at the bottleneck; MHCA is placed at skip connections.

Q3: How did U-Transformer perform compared to U-Net? A: U-Transformer beat U-Net on all organ segmentation tasks, with especially large gains on the pancreas.

Q4: What role does positional encoding play in MHCA? A: Positional encoding is important in MHCA to preserve spatial information when combining features from different levels.

Lecture 6 — Diffusion Models & Stable Diffusion

Q1: What happens in the forward diffusion process? A: Noise is gradually added to the image over T steps via a Markov chain until it becomes pure Gaussian noise.

Q2: Write the reparameterized forward diffusion equation. A: $x_t = \sqrt{\alpha_t} \cdot x_0 + \sqrt{1 - \alpha_t} \cdot \epsilon$, allowing direct jump to any timestep t .

Q3: What does the U-Net predict during reverse diffusion? A: The U-Net predicts the noise added at each step, which is then subtracted to recover the clean image.

Q4: What is the training loss for a diffusion model? A: Simple MSE between true noise and predicted noise: $\| \epsilon - \hat{\epsilon}(x_t, t) \|^2$.

Q5: What is a Latent Diffusion Model (LDM) and why is it useful? A: LDM runs diffusion in VAE-compressed latent space ($48\times$ smaller), making it much more computationally efficient.

Q6: How does Stable Diffusion incorporate text conditioning? A: Text embeddings are injected into the U-Net via cross-attention layers during the denoising process.

Q7: What is Classifier-Free Guidance (CFG)? A: A technique that interpolates between conditional and unconditional predictions using a guidance scale w to control text adherence.

Lecture 7 — Embeddings

Q1: What are the problems with one-hot encoding? A: Large weight matrices, huge dataset needs, high computation/memory cost, and no semantic meaning captured.

Q2: What is the difference between static and contextual embeddings? A: Static embeddings (e.g., Word2Vec) give the same vector regardless of context; contextual embeddings (e.g., BERT) change based on surrounding words.

Q3: How can embeddings be obtained? A: Either via dimensionality reduction (e.g., PCA) or by training them as part of the model end-to-end.

Lecture 8 — BERT & ResNet

Q1: What three types of embeddings are summed to form BERT’s input representation? A: Token embeddings + Segment embeddings + Position embeddings.

Q2: What are BERT’s two pre-training tasks? A: Masked Language Modeling (MLM) with the 80/10/10 masking rule, and Next Sentence Prediction (NSP).

Q3: What is the residual block formula in ResNet and what problem does it solve? A: $y = F(x, W) + x$ — the skip connection solves the vanishing gradient problem in very deep networks.

Q4: How are BERT and ResNet combined for multimodal sentiment analysis? A: BERT processes text and ResNet processes images; their features are combined for sentiment prediction.

Lecture 9 — State Space Models & Mamba

Q1: What are the two core SSM equations? A: $h'(t) = Ah(t) + Bx(t)$ (state update) and $y(t) = Ch(t) + Dx(t)$ (output).

Q2: What is the dual view of SSMs for training vs inference? A: Convolution mode for efficient parallel training; recurrence mode for fast autoregressive inference.

Q3: What makes Mamba’s SSM “selective”? A: In Mamba, matrices B , C , and step size Δ become input-dependent functions, allowing the model to selectively focus on relevant information.

Q4: What are Mamba’s key performance advantages over Transformers? A: $5\times$ throughput, linear scaling with sequence length, and ability to handle million-token sequences.

Lecture 10 — Contrastive Learning & SimCLR

Q1: What is contrastive learning? A: A self-supervised method that learns representations by pulling positive pairs closer and pushing negative pairs apart.

Q2: What is the SimCLR pipeline? A: Augmentation $\rightarrow$ encoder $f(\cdot) \rightarrow$ projection head $g(\cdot) \rightarrow$ NT-Xent loss.

Q3: What are SimCLR’s three key findings? A: Composition of augmentations matters, a non-linear projection head helps, and larger batch sizes improve performance.

Q4: Why is the projection head discarded after SimCLR training? A: The projection head loses information useful for downstream tasks; the encoder representations before it are more general.

Lecture 11 — Transfer Learning & PEFT

Q1: What are the three stages of the modern LLM pipeline? A: Pretraining (language modeling) → Supervised Fine-Tuning (SFT/instruction tuning) → Preference Tuning (RLHF/DPO).

Q2: Why is transfer learning effective (as shown with medical images)? A: A model pretrained on ImageNet reaches 80–90% accuracy on medical images, vs only 60% when trained from scratch.

Q3: What are adapters and where are they inserted? A: Bottleneck modules inserted after self-attention and FFN layers; the original model weights stay frozen.

Lecture 12 — LoRA & QLoRA

Q1: What is the core idea of LoRA? A: Approximate the weight update ΔW as a low-rank decomposition BA (where $r \ll d, k$), drastically reducing trainable parameters.

Q2: How is LoRA initialized and why? A: A is random Gaussian, B is zero — so $BA = 0$ at init, meaning training starts from the original pretrained weights.

Q3: How does LoRA enable multi-task serving? A: Keep one frozen W and swap small (B, A) adapter pairs per task — very low storage and fast switching.

Q4: What is QLoRA and why is it significant? A: QLoRA uses a 4-bit NF4 quantized base model with BF16 LoRA adapters and on-the-fly dequantization — fine-tuning a 7B model in ~5GB memory.

Q5: What is block-wise quantisation? A: Quantising weights in blocks with per-block scaling constants to better handle outlier values and reduce precision loss.

Lecture 13 — RAG (Retrieval-Augmented Generation)

Q1: What is the RAG pipeline? A: Query → embed → vector DB search → dense retrieval → re-rank → augmented prompt → grounded generation → answer with citations.

Q2: How does dense retrieval differ from keyword search? A: Dense retrieval uses text-embedding similarity to capture semantic meaning, not just keyword overlap.

Q3: What is re-ranking in RAG? A: A two-stage process — fast dense retrieval first, then a precise cross-encoder re-scores the top candidates.

Q4: Name three advanced RAG variants. A: Query Rewriting, Multi-query RAG, and Multi-hop RAG.

Q5: What are key RAG evaluation metrics? A: Fluency, Perceived Utility, Citation Recall, Citation Precision, Faithfulness, and Answer Relevance.

Lecture 14 — Flash Attention

Q1: What is the bottleneck in standard attention — compute or memory? A: Memory-bound — the repeated HBM read/writes dominate, not the actual computation.

Q2: What are the two key techniques in Flash Attention? A: Tiling (splitting Q, K, V into blocks processed in SRAM) and recomputation (recomputing S and P in backward pass instead of storing them).

Q3: What is online softmax and why does Flash Attention need it? A: A method using running max $m(x)$ and sum $s(x)$ to compute softmax block-by-block without materializing the full $N \times N$ attention matrix.

Q4: What are Flash Attention's performance gains? A: $3\times$ speedup on GPT-2, 15% end-to-end wall-clock improvement, with exact (not approximate) output.

Lecture 15 — Prompt Engineering

Q1: What do Temperature and Top-p control in LLM generation? A: Temperature controls randomness/creativity; Top-p (nucleus sampling) limits token selection to the smallest set whose cumulative probability exceeds p.

Q2: What are the 7 prompt ingredients? A: Persona, Instruction, Context, Format, Audience, Tone, and Data.

Q3: What is the difference between zero-shot, one-shot, and few-shot prompting? A: Zero-shot gives no examples, one-shot gives one example, few-shot gives multiple examples to guide the model's response.

Q4: What is chain prompting? A: Breaking a complex task into a pipeline of sequential prompts (e.g., Name $\rightarrow$ Slogan $\rightarrow$ Sales Pitch).

Q5: What are the pitfalls of LLM-generated prompts? A: Feedback loops, intent loss, injection bias, and over-optimisation.